

JEE MAIN REVISION MASTER: ATOMIC STRUCTURE 2026 2 QUESTIONS 3.33% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 12 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Atomic Structure - Core Application	12 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Atomic Structure - Core Application (High Yield — 12 PYQs)

Core Principle: Key problem pattern in Kcet Atomic Structure.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Which of the following is the CORRECT statement about Ψ^2 ?..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Which of the following represents de Broglie equation?..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "The number of orbitals associated with ' N ' shell of an atom is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Which of the following is the CORRECT statement about Ψ^2 ?

- (A) Ψ^2 represents atomic orbit
- (B) Probability density of the electron at that point
- (C) $\Psi^2 \neq 0$ for nodes
- (D) Ψ^2 has no physical meaning

Correct Answer: None

Solution Analysis: The correct statement is: Option B: Probability density of the electron at that point. In quantum mechanics, Ψ is the wave function of an electron. The wave function itself, Ψ , does not have direct physical meaning. But the square of the wave function gives the probability density of finding the electron at a particular point. More correctly, we write it as: $|\Psi|^2$. For real wave functions, this is often written simply as: Ψ^2 . So, Ψ^2 tells us the probability density of the electron at that point. Also, at nodes, the probability of finding the electron is zero, so: